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Tutorial Course for Intro to Sensor-Integrated Systems

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Basic Systematic Error

Three Major Assumptions of Ideal Elements:

- **Perfectly linear:** A strictly proportional relationship between input and output.
- **Zero intercept:** When the input is 0, the output is strictly 0 (i.e., $a = 0$).
- **No environmental inputs:** Not affected by corrections or interference from the external environment (e.g., temperature, humidity).

Mathematical Expression: For a single element meeting the above conditions, the input-output equation is very simple:

$$O_i = K_i I_i$$

(Where K_i is the linear sensitivity/slope of the i element)

Derivation Process: Signals are passed like a relay race; the output of the previous element is the input for the next.

•Level 1 Output: $O_1 = K_1 I$

•Level 2 Output: $O_2 = K_2 O_1 = K_2 K_1 I$

•Level 3 Output: $O_3 = K_3 O_2 = K_3 K_2 K_1 I$

Conclusion: For an entire measurement system containing n elements, the final system output O is:

$$O = (K_1 K_2 K_3 \dots K_n) I$$

Core Summary: The **Total Sensitivity** of a series system, K_{sys} , is equal to the **product** of the individual element sensitivities

$$K_{sys} = \prod_{i=1}^n K_i = K_1 K_2 \dots K_n$$

Error Definition: Measurement error E is defined as the difference between the measured value (final system output O) and the true value (initial system input I).

$$E = O - I$$

$$E = (K_{sys} - 1)I$$

Exercise

A force measurement system consists of four elements with sensitivities 10^{-2} , 5×10^{-2} , 10^3 and 1.9. Find the system error for a true value input of 10 kN.

$$K_{sys} = K_1 \times K_2 \times K_3 \times K_4$$

$$K_{sys} = 0.95$$

$$E = (K_{sys} - 1) \times I$$

$$E = (0.95 - 1) \times 10$$

$$E = -0.05 \times 10$$

$$E = -0.5 \text{ kN}$$

A level measurement system consists of three linear elements with sensitivities of 0.050, 21.5 and 0.99. Find the system error for a true value input of 5.0 metres.

$$K_{sys} = K_1 \times K_2 \times K_3$$

$$K_{sys} = 1.06425$$

$$E = (K_{sys} - 1) \times I$$

$$E = (1.06425 - 1) \times 5.0 \text{ m}$$

$$E = 0.06425 \times 5.0 \text{ m}$$

$$E = 0.32125 \text{ m}$$

Error Band Modeling and Propagation Analysis

1. Introduction: What is an "Error Band" ($\pm h$)?

Engineering Reality: Sensor manufacturers usually only provide an "accuracy range" (e.g., $\pm 1\%$). This means the actual output is restricted within a band-shaped region fluctuating above and below the ideal linear path, with a total width of $2h$.

2. Core Transformation: Turning "Range" into "Variance"

- **Statistical Assumption:** Assume the probability of the error falling at any point within $\pm h$ is equal; that is, it follows a **rectangular distribution** (uniform distribution).
- An error band with width $2h$ is equivalent to an internal noise source within the system, with its own variance defined as:

$$\sigma_{own}^2 = \frac{h^2}{3}$$

3. Variance Propagation Rules for Series Systems

- In a series system, errors accumulate level by level like a rolling snowball and are amplified by the sensitivity (K) of each element.
- **Recursive Formula:**

$$\sigma_{out}^2 = (K^2 \times \sigma_{in}^2) + \frac{h^2}{3}$$

4. Key Points

- The initial input of the system (the true physical quantity) usually has no random error, therefore $\sigma_{in1}^2 = 0$.
- **The sensitivity K_1 of the first-stage element will never amplify its own error!** The output variance of the first stage is simply $\frac{h_1^2}{3}$.

Exercise

A pressure measurement system consists of a pressure sensor, deflection bridge, amplifier and recorder. Table Prob. 2 gives the linear sensitivities and error bandwidths for each element in the system.

- (a) Calculate the standard deviation σ_E of the error distribution function.
- (b) Given that the recorder is incorrectly adjusted so that its sensitivity is 225 Pa mV^{-1} , calculate the mean error $\bar{E}$ for an input pressure of $5 \times 10^3 \text{ Pa}$.

Element	Linear sensitivity K	Error bandwidth $\pm h$
Pressure sensor	$10^{-4} \Omega \text{ Pa}^{-1}$	$\pm 0.005 \Omega$
Deflection bridge	$4 \times 10^{-2} \text{ mV } \Omega^{-1}$	$\pm 5 \times 10^{-4} \text{ mV}$
Amplifier	10^3 mV mV^{-1}	$\pm 0.5 \text{ mV}$
Recorder	250 Pa mV^{-1}	$\pm 100 \text{ Pa}$

Exercise

(a) Calculate the standard deviation σ_E of the error distribution function.

$$\sigma_1^2 = K_1^2 \cdot (0) + \frac{h_1^2}{3} = \frac{\mathbf{h_1^2}}{3}$$

$$\sigma_1^2 = \frac{0.005^2}{3} = \frac{25 \times 10^{-6}}{3}$$

$$\sigma_2^2 = K_2^2 \cdot (\sigma_1^2) + \frac{h_2^2}{3} = \mathbf{K_2^2 \frac{h_1^2}{3} + \frac{h_2^2}{3}}$$

$$\sigma_2^2 = (4 \times 10^{-2})^2 \times \sigma_1^2 + \frac{(5 \times 10^{-4})^2}{3} = \frac{\mathbf{29 \times 10^{-8}}}{3}$$

$$\sigma_3^2 = K_3^2 \cdot (\sigma_2^2) + \frac{h_3^2}{3} = K_3^2 \left(K_2^2 \frac{h_1^2}{3} + \frac{h_2^2}{3} \right) + \frac{h_3^2}{3} = \mathbf{K_3^2 K_2^2 \frac{h_1^2}{3} + K_3^2 \frac{h_2^2}{3} + \frac{h_3^2}{3}} \quad \sigma_3^2 = (10^3)^2 \times \sigma_2^2 + \frac{0.5^2}{3} = \frac{\mathbf{0.54}}{3} = \mathbf{0.18}$$

$$\sigma_E^2 = \sigma_4^2 = K_4^2 \cdot (\sigma_3^2) + \frac{h_4^2}{3} = \mathbf{K_4^2 K_3^2 K_2^2 \frac{h_1^2}{3} + K_4^2 K_3^2 \frac{h_2^2}{3} + K_4^2 \frac{h_3^2}{3} + \frac{h_4^2}{3}}$$

$$\sigma_E^2 = (250)^2 \times \sigma_3^2 + \frac{100^2}{3} \approx \mathbf{14583.33}$$

$$\sigma_E = \sqrt{14583.33} \approx \mathbf{120.76 \text{ Pa}}$$

(b) Given that the recorder is incorrectly adjusted so that its sensitivity is 225 Pa mV^{-1} , calculate the mean error $\bar{E}$ for an input pressure of $5 \times 10^3 \text{ Pa}$.

$$K_{\text{actual}} = K_1 \times K_2 \times K_3 \times K_4'$$

$$\bar{O} = K_{\text{actual}} \times \bar{I} = 0.9 \times (5 \times 10^3 \text{ Pa}) = 4500 \text{ Pa}$$

$$K_{\text{actual}} = 4 \times 10^{-3} \times 225 = 0.9$$

$$\bar{E} = \bar{O} - \bar{I}$$

$$\bar{E} = 4500 \text{ Pa} - 5000 \text{ Pa} = -500 \text{ Pa}$$

High-Gain Negative Feedback & Error Reduction

1. The Pain Point: Why are Open-Loop Systems "Fragile"?

In a series (open-loop) system, a parameter drift in **any single element** of the forward path (e.g., an amplifier affected by temperature, or a spring aging) will be directly amplified proportionally, leading to massive measurement errors. Searching for "perfect elements" is prohibitively expensive.

2. The Solution: Introducing a Negative Feedback Structure

Core Idea: Instead of blindly trusting the processing result, "pull back" the **actual output** and compare it with the **intended input** through subtraction. Use the resulting **"error signal"** to drive the system to self-correct.

3. Closed-Loop Transfer Function

The overall sensitivity of a closed-loop system is no longer a simple product of elements, but rather:

$$K_{\text{closed}} = \frac{K_{\text{forward}}}{1 + K_{\text{forward}} \cdot K_{\text{feedback}}}$$

4. High-Gain Approximation (High-Gain Magic)

If we intentionally make the gain of the forward amplifier massive, such that the loop gain $K_{\text{forward}} \cdot K_{\text{feedback}} \gg 1$, the "1" in the denominator becomes negligible, and a wonderful simplification occurs:

$$K_{\text{closed}} \approx \frac{K_{\text{forward}}}{K_{\text{forward}} \cdot K_{\text{feedback}}} = \frac{1}{K_{\text{feedback}}}$$

5. Engineering Takeaway

The "Leverage" Effect: The overall accuracy of the system becomes **completely decoupled from the forward path** (which can now use cheap, drift-prone components) and depends **solely on the stability of the single element in the feedback loop!** This drastically improves the system's immunity to interference.

Exercise

Figure Prob. 3 shows a block diagram of a force transducer using negative feedback. The elastic sensor gives a displacement output for a force input; the displacement sensor gives a voltage output for a displacement input. V_S is the supply voltage for the displacement sensor.

(a) Calculate the output voltage V_0 when

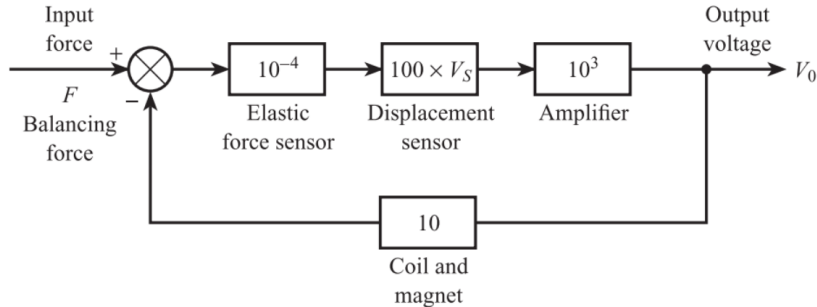
(i) $V_S = 1.0 \text{ V}$, $F = 50 \text{ N}$

$V_0 \approx 4.9505 \text{ V}$

(ii) $V_S = 1.5 \text{ V}$, $F = 50 \text{ N}$.

$V_0 \approx 4.9669 \text{ V}$

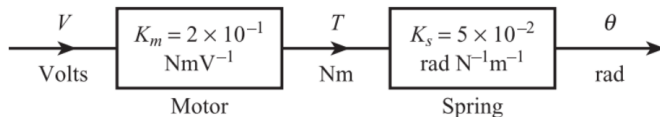
(b) Comment on the practical significance of the variation of the supply voltage V_S .



Exercise

Figure Prob. 4 is a block diagram of a voltmeter. The motor produces a torque T proportional to voltage V , and the output angular displacement θ of the spring is proportional to T . The stiffness K_s of the spring can vary by $\pm 10\%$ about the nominal value of $5 \times 10^{-2} \text{ rad N}^{-1}\text{m}^{-1}$. Given that the following are available:

- (i) a d.c. voltage amplifier of gain 1000
 - (ii) a voltage subtraction unit
 - (iii) a stable angular displacement transducer of sensitivity 100 V rad^{-1} ,
- (a) draw a block diagram of a modified system using these components, which reduces the effect of changes in K_s ;
- (b) calculate the effect of a 10% increase in K_s on the sensitivity of the **modified** system.



- Calculate the Final Sensitivity Variation Rate (Impact):

$$\text{Variation Rate} = \left(\frac{11011}{11010} - 1 \right) \times 100\% = \frac{1}{11010} \times 100\% \approx 0.009\%$$

Comprehensive Statistical Error Analysis for Non-Ideal Systems

1. Real-World Complexity

Real-world measurement systems are typically **non-linear** (including quadratic terms, exponential terms, etc.) and are simultaneously affected by cross-interference from **multiple random variables** (e.g., ambient temperature, fluid density, component tolerances).

2. The "Dual-Track" Algorithm

When solving problems, calculations for the **mean value** and the **fluctuations (variance)** must be kept strictly separate and treated as independent tracks:

Track 1: Calculating Mean Error $\bar{E}$ (Systematic Bias)

- ✓ **Rule:** Ignore fluctuations and plug in mean values. Take all nominal mean values provided in the problem and substitute them step-by-step into the complex non-linear equations to calculate the final measured mean O .
- ✓ **Formula:** System Error $\bar{E} = \bar{O} - \text{True Input } I$.

Track 2: Calculating Error Standard Deviation σ_E (Random Fluctuation)

- ✓ **Rule:** Partial derivatives act as weights for amplification. Use the Law of Error Propagation for multi-variable functions to calculate the accumulation of variance:

$$\sigma_{out}^2 = \left(\frac{\partial O_{out}}{\partial x_1} \right)^2 \sigma_{x_1}^2 + \left(\frac{\partial O_{out}}{\partial x_2} \right)^2 \sigma_{x_2}^2 + \dots$$

- ✓ **Critical Pitfall:** The partial derivatives must be evaluated at the **mean values** derived in "Track 1"! After calculating the total variance, take the square root to find σ_E .

3. Engineering Diagnosis & Optimization

- ✓ **$\bar{E}$ is large $\rightarrow$ Modify the software algorithm.** The inverse calculation formula in the microcontroller is inaccurate (e.g., truncation errors). Upgrade to a more precise polynomial inverse equation or use a lookup table.
- ✓ **σ_E is large $\rightarrow$ Increase dynamic compensation.** Identify the term in the variance calculation with the largest value (usually environmental interference). Add auxiliary sensors (like thermometers or barometers) to the system to feed real-time environmental data into the microcontroller for active compensation/cancellation.

Exercise

A measurement system consists of a chromel–alumel thermocouple (with cold junction compensation), a millivolt-to-current converter and a recorder. Table Prob. 1 gives the model equations, and parameters for each element. Assuming that all probability distributions are normal, calculate the mean and standard deviation of the error probability distribution, when the input temperature is 117 °C.

	Chromel–alumel thermocouple	e.m.f.-to-current converter	Recorder
Model equation	$E = C_0 + C_1T + C_2T^2$	$i = K_1E + K_ME\Delta T_a + K_I\Delta T_a + a_1$	$T_M = K_2i + a_2$
Mean values	$\bar{C}_0 = 0.00$ $\bar{C}_1 = 4.017 \times 10^{-2}$ $\bar{C}_2 = 4.66 \times 10^{-6}$	$\bar{K}_1 = 3.893$ $\Delta T_a = -10$ $\bar{a}_1 = -3.864$ $\bar{K}_M = 1.95 \times 10^{-4}$ $\bar{K}_I = 2.00 \times 10^{-3}$	$\bar{K}_2 = 6.25$ $\bar{a}_2 = 25.0$
Standard deviations	$\sigma_{C_0} = 6.93 \times 10^{-2}$ $\sigma_{C_1} = \sigma_{C_2} = 0$	$\sigma_{a_1} = 0.14, \sigma_{\Delta T_a} = 10$ $\sigma_{K_1} = \sigma_{K_M} = \sigma_{K_I} = 0$	$\sigma_{a_2} = 0.30$ $\sigma_{K_2} = 0.0$

Exercise

Calculating the Mean Measured Value and Mean System Error $\bar{E}_{sys}$

$$\bar{E} = \bar{C}_0 + \bar{C}_1 T + \bar{C}_2 T^2 = 4.76368 \text{ mV}$$

$$\bar{i} = \bar{K}_1 \bar{E} + \bar{K}_M \bar{E} \overline{\Delta T_a} + \bar{K}_I \overline{\Delta T_a} + \bar{a}_1 = 14.6517 \text{ mA}$$

$$\bar{T}_M = \bar{K}_2 \bar{i} + \bar{a}_2 = 116.573 \text{ }^\circ\text{C}$$

$$\bar{E}_{sys} = \bar{T}_M - T = 116.573 - 117 = -\mathbf{0.427} \text{ }^\circ\text{C}$$

Calculating Error Variance and Standard Deviation $\sigma_{E_{sys}}$

$$\sigma_E^2 = \left(\frac{\partial E}{\partial C_0} \right)^2 \sigma_{C_0}^2 = 0.004802 \text{ mV}^2$$

$$\sigma_i^2 = \left(\frac{\partial i}{\partial E} \right)^2 \sigma_E^2 + \left(\frac{\partial i}{\partial \Delta T_a} \right)^2 \sigma_{\Delta T_a}^2 + \left(\frac{\partial i}{\partial a_1} \right)^2 \sigma_{a_1}^2 = 0.09316 \text{ mA}^2$$

$$\sigma_{T_M}^2 = \left(\frac{\partial T_M}{\partial i} \right)^2 \sigma_i^2 + \left(\frac{\partial T_M}{\partial a_2} \right)^2 \sigma_{a_2}^2 = 3.7291 \text{ (}^\circ\text{C)}^2$$

$$\sigma_{E_{sys}} = \sqrt{\sigma_{T_M}^2} = \sqrt{3.7291} \approx \mathbf{1.931} \text{ }^\circ\text{C}$$

Exercise

A fluid velocity measurement system consists of a pitot tube, a differential pressure transmitter, an 8-bit analogue-to-digital converter and a microcontroller with display facilities. Table Prob. 6 gives the model equations and parameters for each element in the system. The microcontroller calculates the measured value of velocity assuming a constant density.

- (a) Estimate the mean and standard deviation of the error probability density function assuming the true value of velocity v_T is 14.0 m s^{-1} . Use the procedure of Table 3.1, treating the rectangular distributions as normal with $\sigma = h/\sqrt{3}$.
- (b) Explain briefly what modifications could be made to the system to reduce the quantities calculated in (a).

	Pitot tube	Differential pressure transmitter	Analogue-to-digital converter	Microcontroller with display
Model equations	$\Delta P = \frac{1}{2}\rho v_T^2$	$i = K_1\Delta P + a_1$	$n = K_2i + a_2$	$v_M = K_3\sqrt{n - 51}$
Mean values	$\bar{\rho} = 1.2$	$\bar{K}_1 = 0.064$ $\bar{a}_1 = 4.0$	$\bar{K}_2 = 12.80$ $\bar{a}_2 = 0.0$ n rounded off to nearest integer	$\bar{K}_3 = 1.430$
Half-widths of rectangular distribution	$h_\rho = 0.1$	$h_{a_1} = 0.04$	$h_{a_2} = 0.5$	$h_{K_3} = 0.0$

Exercise

- (a) Estimate the mean and standard deviation of the error probability density function assuming the true value of velocity v_T is 14.0 m s^{-1} . Use the procedure of Table 3.1, treating the rectangular distributions as normal with $\sigma = h/\sqrt{3}$.

Part (a): Statistical Analysis

I. Mean Systematic Error $\bar{E}$ (Nominal Tracking)

1. Core Formulas (Substitute Mean Values):

- Pitot tube:** $\Delta\bar{P} = 0.5\bar{\rho}v_T^2 = 117.6 \text{ Pa}$
- Transmitter:** $\bar{i} = \bar{K}_1\Delta\bar{P} + \bar{a}_1 = 11.5264 \text{ mA}$
- ADC:** $\bar{n} = \bar{K}_2\bar{i} + \bar{a}_2 = 147.5379$
- Microcontroller:** $\bar{v}_M = \bar{K}_3\sqrt{\bar{n} - 51} = 14.050 \text{ m/s}$

2. Final Result:

$$\bar{E} = \bar{v}_M - v_T = 14.050 - 14.0 = +\mathbf{0.050 \text{ m/s}}$$

II. Error Standard Deviation σ_E (Variance Propagation)

1. Core Formulas (Partial Derivatives Squared):

- Pitot tube variance:**

$$\sigma_{\Delta P}^2 = \left(\frac{\partial \Delta P}{\partial \rho}\right)^2 \sigma_{\rho}^2 = (98)^2 \times 0.003333 = 32.0133 \text{ Pa}^2$$

- Transmitter variance:**

$$\sigma_i^2 = \left(\frac{\partial i}{\partial \Delta P}\right)^2 \sigma_{\Delta P}^2 + \left(\frac{\partial i}{\partial a_1}\right)^2 \sigma_{a_1}^2 = 0.13166 \text{ mA}^2$$

- ADC variance:**

$$\sigma_n^2 = \left(\frac{\partial n}{\partial i}\right)^2 \sigma_i^2 + \left(\frac{\partial n}{\partial a_2}\right)^2 \sigma_{a_2}^2 = 21.654$$

- Microcontroller total variance:** (Note: derivative evaluated at $\bar{n}$)

$$\sigma_{v_M}^2 = \left(\frac{\partial v_M}{\partial n}\right)^2 \sigma_n^2 = (0.07277)^2 \times 21.654 = 0.1147 \text{ (m/s)}^2$$

2. Final Result:

$$\sigma_E = \sqrt{\sigma_{v_M}^2} = \sqrt{0.1147} \approx \mathbf{0.339 \text{ m/s}}$$

Exercise

- (b) Explain briefly what modifications could be made to the system to reduce the quantities calculated in (a).

Why is there a systematic mean error of $+0.050 \text{ m s}^{-1}$?

No matter how the system measures, the indicated value on the meter is always 0.050 m s^{-1} higher than the true physical velocity. This is a fixed **Systematic Bias**.

In modern measurement systems, this type of pure mathematical deviation is almost always caused by an **inaccurate inverse equation** programmed into the microcontroller. To debug this, we need to derive the "perfect theoretical formula" ourselves:

- **Forward Model (Physical World $\rightarrow$ Digital Signal):**

We multiply the nominal equations of the first three stages (Pitot tube, transmitter, ADC) together:

$$\Delta P = 0.5 \times 1.2 \times v^2 = 0.6v^2$$

$$i = 0.064\Delta P + 4.0 = 0.0384v^2 + 4.0$$

$$n = 12.8i = 12.8(0.0384v^2 + 4.0) = \mathbf{0.49152v^2 + 51.2}$$

Optimization Plan: Modify the microcontroller's software code.

There is no need to replace any expensive hardware. By simply updating the firmware to use the precise formula $v = 1.4264\sqrt{n - 51.2}$, this systematic error will be completely eliminated.

Exercise

- (b) Explain briefly what modifications could be made to the system to reduce the quantities calculated in (a).

Why is there a random fluctuation (standard deviation) of 0.339 m s^{-1} ?

Even if the true fluid velocity in the pipe is firmly fixed at 14.0 m s^{-1} , the meter's reading will continuously jump around 14.0 m s^{-1} , with a standard deviation of 0.339 m s^{-1} . This represents a **Random Error**.

- **ADC's internal quantization variance:** $\sigma_{a_2}^2 \approx 0.083$
- **Transmitter's internal bias variance:** $\sigma_{a_1}^2 \approx 0.0005$
- **Variance propagated from the fluid density ρ fluctuation in the Pitot tube:** $\sigma_{\Delta P}^2 \approx$

32.0133

Optimization Plan:

In real-world engineering, the density ρ of a fluid (like air or water) is never a constant. It fluctuates drastically with the real-time temperature and absolute pressure inside the pipe. Hard-coding $\bar{\rho}$ as a constant 1.2 in the system design is a severe flaw.

Add dynamic compensation hardware. We must install an additional temperature sensor and an absolute pressure sensor into the pipeline. The real-time signals from these sensors should be fed into the microcontroller. Using the ideal gas law ($\rho = P/RT$), the microcontroller can compute the true, real-time density ρ on the fly and feed it into the velocity formula. In the industry, this is known as "Temperature and Pressure Compensation" (or Dynamic Compensation).

Exercise

A temperature measurement system consists of a thermistor, bridge and recorder. Table Prob. 7 gives the model equations, mean values and standard deviations for each element in the system. Assuming all probability distributions are normal, calculate the mean and standard deviation of the error probability density function for a true input temperature of 320 K.

	Thermistor	Bridge	Recorder
Model equations	$R_\theta = K_1 \exp\left(\frac{\beta}{\theta}\right)$	$V_0 = V_s \left\{ \frac{1}{1 + \frac{3.3}{R_\theta}} - a_1 \right\}$	$\theta_M = K_2 V_0 + a_2$
Mean values	$\bar{K}_1 = 5 \times 10^{-4} \text{ k}\Omega$ $\bar{\beta} = 3 \times 10^3 \text{ K}$	$\bar{V}_s = -3.00 \text{ V}$ $\bar{a}_1 = 0.77$	$\bar{K}_2 = 50.0 \text{ K/V}$ $\bar{a}_2 = 300 \text{ K}$
Standard deviations	$\sigma_{K_1} = 0.5 \times 10^{-4}$ $\sigma_\beta = 0$	$\sigma_{V_s} = 0.03$ $\sigma_{a_1} = 0.01$	$\sigma_{K_2} = 0.0$ $\sigma_{a_2} = 3.0$

Exercise

Part (a): Statistical Analysis

I. Mean Systematic Error $\bar{E}$ (Nominal Tracking)

1. Core Formulas (Substitute Mean Values):

- **Thermistor:** $\bar{R}_\theta = \bar{K}_1 \exp\left(\frac{\bar{\beta}}{\theta}\right) = 5.895 \text{ k}\Omega$
- **Bridge:** $\bar{V}_0 = \bar{V}_S \left\{ \frac{1}{1+3.3/\bar{R}_\theta} - \bar{a}_1 \right\} = 0.3867 \text{ V}$
- **Recorder:** $\bar{\theta}_M = \bar{K}_2 \bar{V}_0 + \bar{a}_2 = 319.335 \text{ K}$

2. Final Result:

$$\bar{E} = \bar{\theta}_M - \theta = 319.335 - 320 = -\mathbf{0.665 \text{ K}}$$

II. Error Standard Deviation σ_E (Variance Propagation)

1. Core Formulas (Partial Derivatives Squared):

• Thermistor variance:

$$\sigma_{R_\theta}^2 = \left(\frac{\partial R_\theta}{\partial K_1} \right)^2 \sigma_{K_1}^2 = 0.3475 \text{ (k}\Omega)^2$$

• Bridge variance: (Note: This is the most error-prone derivative step)

$$\sigma_{V_0}^2 = \left(\frac{\partial V_0}{\partial V_S} \right)^2 \sigma_{V_S}^2 + \left(\frac{\partial V_0}{\partial a_1} \right)^2 \sigma_{a_1}^2 + \left(\frac{\partial V_0}{\partial R_\theta} \right)^2 \sigma_{R_\theta}^2 = 0.00568 \text{ V}^2$$

• Recorder (Total System) variance:

$$\sigma_{\theta_M}^2 = \left(\frac{\partial \theta_M}{\partial V_0} \right)^2 \sigma_{V_0}^2 + \left(\frac{\partial \theta_M}{\partial a_2} \right)^2 \sigma_{a_2}^2 = 23.2 \text{ K}^2$$

2. Final Result:

$$\sigma_E = \sqrt{\sigma_{\theta_M}^2} = \sqrt{23.2} \approx \mathbf{4.82 \text{ K}}$$